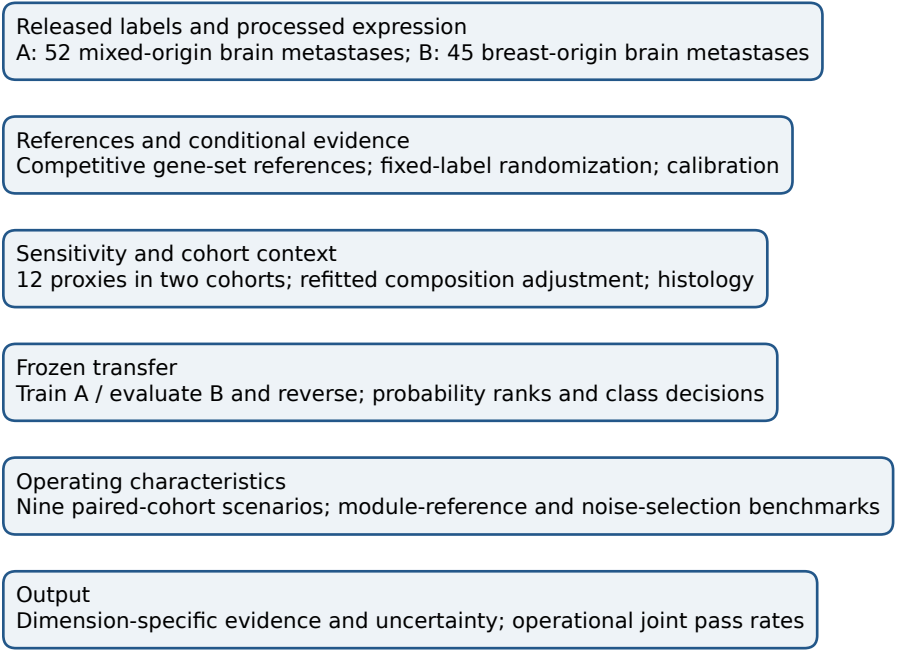
Released labels and processed expression
A: 52 mixed-origin brain metastases; B: 45 breast-origin brain metastases
References and conditional evidence
Competitive gene-set references; fixed-label randomization; calibration
Sensitivity and cohort context
12 proxies in two cohorts; refitted composition adjustment; histology
Frozen transfer
Train A / evaluate B and reverse; probability ranks and class decisions
Operating characteristics
Nine paired-cohort scenarios; module-reference and noise-selection benchmarks
Output
Dimension-specific evidence and uncertainty; operational joint pass rates